ENGINE PERFORMANCE & ECONOMICS REPORT

Human-Equivalent Workload Analysis Across the Sophon Finance Systems Engine Portfolio

PREPARED BY Sophon Narith — Founder, Sophon Finance Systems

CONTACT contact@sophonfinance.com · sophonfinance.com · linkedin.com/in/sophonnarith · github.com/sophonfinance-wq

SUBJECT Measured engine throughput vs. estimated professional labor for the same scope of finance and tax work

BASIS Engine figures: observed runtimes and shipped test suites. Human figures: standard engagement budgeting for the equivalent scope, stated as estimates. Cost basis: $100/hr fully-loaded internal; $250/hr market-billed.

STATUS **FOR CLIENT DISTRIBUTION**

1.0 PURPOSE AND METHOD

This report answers one question per engine: **what does the same scope of work cost when a person does it, versus when the engine does it and a person reviews it?** Eight production engines are covered. Every engine is public, runnable, and tested — the code can be inspected at github.com/sophonfinance-wq. Engine run figures are observed. Human-hour baselines are engineering estimates built the way an engagement is budgeted, and are deliberately conservative: they assume a competent senior preparer, no interruptions, and no rework. Where this report references field results, the engagement details are generalized; client work is confidential.

Every engine operates under the same control model: the engine drafts, an independent pass verifies, and a person holds final sign-off. Nothing posts, files, or sends itself.

2.0 EXECUTIVE SUMMARY — THE PORTFOLIO AT A GLANCE

ENGINE	BENCHMARK UNIT	HUMAN HRS*	ENGINE + REVIEW	SPEEDUP	SAVED / UNIT @$100
Tax Surplus (T1134)	per foreign affiliate, full-history surplus & ACB	24–40	0.1 + 2.0	**12–19×**	$2,190–$3,790
Partnership 1065	per partnership, K-1s + §704(c) support	12–20	0.1 + 1.5	**8–13×**	$1,040–$1,840
Month-End Close	per monthly close, multi-entity	30–50	0.3 + 4.0	**7–12×**	$2,570–$4,570
Cash Reconciliation	per monthly cash/debt recon, multi-region	12–20	0.2 + 1.5	**7–12×**	$1,030–$1,830
Validation Engine	per workbook tie-out / control test	6–10	0.1 + 0.5	**10–17×**	$540–$940
Triangulate (AI QC)	per independent re-review of a deliverable	8–12	0.3 + 0.8	**7–11×**	$690–$1,090
Knowledge Brain	per meeting ingested + per cited retrieval	3–5	0.2 + 0.1	**10–17×**	$270–$470
Finance Atlas	per department system map, built + refreshed	20–30	0.2 + 1.0	**17–25×**	$1,880–$2,880

*Standard engagement estimate for the identical scope — senior preparer, uninterrupted, no rework. Engine+review = observed run time plus human review and approval of the output.

ILLUSTRATIVE ANNUAL PROFILE — a mid-size multi-entity group		
12 monthly closes · 12 cash recons · 8 partnerships · 4 affiliates (T1134) · 24 validation runs · 12 QC reviews	**Human: ~1,150 hrs**	**Engine + review: ~110 hrs**
Estimated annual labor value released at $100/hr internal — before billed-rate arbitrage at $250/hr	**≈ $104,000 / yr**	≈ 10.4× ROI on review time

3.0 ENGINE PERFORMANCE SHEETS

E-01 TAX SURPLUS ENGINE

Canadian foreign-affiliate surplus pools & ACB (CRA Form T1134)

Maintains exempt / taxable / pre-acquisition surplus pools and adjusted cost base year-by-year across a multi-tier cross-border structure, with every figure traceable to its rule (Reg. 5907) and source line.

BENCHMARK	One affiliate, full-history pools + ACB, review-ready workpaper
MANUAL REALITY	3–6 hrs per affiliate-year compounding across a decade of history; one FX slip poisons every later year
HUMAN ESTIMATE	24–40 hrs per affiliate
ENGINE OBSERVED	Full recompute in seconds; deterministic; 2 hrs human review
PER-UNIT ECONOMICS	$2,400–$4,000 manual vs ~$210 engine+review — saves $2,190–$3,790 per affiliate

■ The compounding-error class disappears: the engine recomputes the entire history every run, so a corrected prior year re-flows automatically instead of by hand.

E-02 PARTNERSHIP 1065 AUTOMATION

Form 1065 / Schedule K-1 support with IRC §704(c)

Builds the book-to-tax bridge, allocates Schedule K items to partners with penny control, carries §704(c) built-in-gain layers (traditional method / ceiling rule), and emits a review-ready support package.

BENCHMARK	One partnership, full 1065 support + every K-1, allocation schedules tied
MANUAL REALITY	Allocation drift and penny breakage across partners is the classic late-night failure
HUMAN ESTIMATE	12–20 hrs per partnership
ENGINE OBSERVED	Package in under a minute; 1.5 hrs review
PER-UNIT ECONOMICS	$1,200–$2,000 manual vs ~$160 — saves $1,040–$1,840 per partnership

■ Penny control is enforced by construction — allocations must tie to the ledger or the run refuses to emit.

E-03 MONTH-END CLOSE AUTOMATION

Multi-entity recurring close on multiple accounting databases

Drafts the recurring journal rhythm — prepaid amortization, depreciation, deferred rent/CAM, accrued fees, intercompany interest, allocations — under ten deterministic controls, refusing any entry that does not tie.

BENCHMARK	One monthly close across a multi-entity group
MANUAL REALITY	The same 8 entry families rebuilt by hand every month; schedule-to-GL ties eat the calendar
HUMAN ESTIMATE	30–50 hrs per close
ENGINE OBSERVED	Draft entries + tie-outs in minutes; 4 hrs controller review
PER-UNIT ECONOMICS	$3,000–$5,000 manual vs ~$430 — saves $2,570–$4,570 per month, every month

■ Out-of-tie entries cannot post. The close package arrives drafted, balanced, and documented.

E-04 CASH & DEBT RECONCILIATION

Materiality-based recon with an audit-ready evidence log

Ties cash and debt to the ledger across regions and banks the same way every month, writing a structured evidence log of what it read, matched, and flagged.

BENCHMARK	One monthly multi-region cash + debt reconciliation
MANUAL REALITY	Bank migrations, shared tabs, and off-ledger successor accounts hide real balances
HUMAN ESTIMATE	12–20 hrs per month
ENGINE OBSERVED	Recon pass in minutes; 1.5 hrs review of exceptions
PER-UNIT ECONOMICS	$1,200–$2,000 manual vs ~$170 — saves $1,030–$1,830 per month

■ Completeness sweeps beat spot-checks: every source balance must land somewhere or it is flagged — zero unaccounted.

E-05 VALIDATION ENGINE

Deterministic, read-only rule registry over finished workbooks

Runs a registry of tie-out and control rules over .xlsx workbooks and their JSON exports, flagging exceptions without ever mutating the source — the checker cannot corrupt what it checks.

BENCHMARK	One workbook package tie-out / pre-sign-off control test
MANUAL REALITY	Manual ticking finds what it looks at; silence about the rest
HUMAN ESTIMATE	6–10 hrs per package
ENGINE OBSERVED	Full rule sweep in seconds; 0.5 hr review of exceptions
PER-UNIT ECONOMICS	$600–$1,000 manual vs ~$60 — saves $540–$940 per package

■ Read-only by design and repeatable on every save — the recalculation gate that caught real defects in field use.

E-06 TRIANGULATE — AI VALIDATION FRAMEWORK

Separation of duties for AI-assisted workpapers

Preparer, reviewer, specialist, deterministic audit, and human gate as distinct roles. Findings are hypotheses until a source document confirms them; the reconciler rebuilds from source rather than diffing prior conclusions.

BENCHMARK	One independent re-review of a finished deliverable
MANUAL REALITY	A true second preparer rarely exists; review inherits the preparer's blind spots
HUMAN ESTIMATE	8–12 hrs (second preparer + manager pass)
ENGINE OBSERVED	Multi-role review ~1 hr including human reconciliation
PER-UNIT ECONOMICS	$800–$1,200 manual vs ~$110 — saves $690–$1,090 per deliverable

■ Field-proven: on a live 150-entity schedule, two independent AI reviewers and the preparer agreed — and were wrong. The source-first pass caught an approver's answer recorded outside the expected column, and separately overturned a plausible finding by opening the signed filing. Consensus is not verification.

E-07 KNOWLEDGE BRAIN ENGINE

Citation-governed knowledge base from recorded meetings

Turns engagement meetings into a queryable base that answers with the verbatim decision, dated and timestamped — used before meetings to prepare and during workpapers to cite prior decisions word-for-word.

BENCHMARK	One meeting ingested; one cited retrieval into a workpaper
MANUAL REALITY	Decisions live in memory and old notebooks; re-litigating them costs meetings
HUMAN ESTIMATE	3–5 hrs (minute-taking + later archaeology per retrieval)
ENGINE OBSERVED	Ingest ~10 min; retrieval with citation in seconds
PER-UNIT ECONOMICS	$300–$500 manual vs ~$30 — saves $270–$470 per meeting cycle

■ The citation discipline is the point: a quote with a date and timestamp ends arguments that summaries start.

E-08 FINANCE OPERATIONS ATLAS

Interactive single-file map of a finance department

Generates a clickable HTML atlas of drives, workstreams, recurring calendar, and a find-it directory from a Python data model — the onboarding and continuity document nobody has time to write.

BENCHMARK	One department atlas, built and kept current
MANUAL REALITY	Tribal knowledge; the map lives in one person's head and leaves with them
HUMAN ESTIMATE	20–30 hrs to author; decays immediately
ENGINE OBSERVED	Generated in minutes from the data model; 1 hr curation
PER-UNIT ECONOMICS	$2,000–$3,000 manual vs ~$120 — saves $1,880–$2,880 per build/refresh

■ Because it is generated, it is regenerable — the map stays true as the department changes.

4.0 FIELD CASE — A LIVE 150-ENTITY TAX SCHEDULE, MEASURED

The clearest evidence is a real engagement. In July 2026 the full stack — document-sweep tooling, the validation engine, and Triangulate — prepared the annual tax-return status schedule for a private multi-entity real-estate group (~150 filing entities, four evidence sources, three independent review rounds). Details are generalized; the client's work is confidential. The machine time below is measured from the engagement log; the human column is the same scope budgeted the way any firm budgets it.

PHASE	SCOPE	HUMAN EST.	MEASURED
Records sweep	146 entity folders, ~50 filed returns opened and read	20 h	**1.5 h**
Registry reconciliation	271 legal-entity records × 3 tabs vs 159 schedule rows	10 h	**0.7 h**
Roll-forward + build	Schedule roll, 43-item decision list, final build w/ formula rebuild	17 h	**1.7 h**
Independent review ×3	Adversarial pass, second AI vendor, source-first reconciliation	14 h	**1.2 h**
Evidence, comms, docs	Org-chart pass, disputed-entity reads, briefings	14.5 h	**1.5 h**
TOTAL	assigned one morning — verified final the next evening	**~75 h / 3–4 wks**	**6.6 h / ~30 hrs elapsed**

Outcome: zero formula errors on full recalculation; every removal cited to the actual filed return; three completeness gaps surfaced that spot-checking would have missed; and one approver decision — recorded outside the expected answer column — recovered by the source-first review after both prior review passes missed it. The deliverable shipped roughly a month ahead of its traditional deadline.

5.0 WHY THE NUMBERS HOLD — CONTROLS & VERIFICATION

- Public, runnable code — every engine demonstrable end-to-end on fictional seeded data; 13,000+ automated tests gate CI.
- Evidence first — computed figures carry citations to file, sheet, and row. No orphan numbers.
- Independent verification — a separate pass with separate logic recomputes the work; the preparer never grades itself.
- Human gate — nothing posts, files, or sends itself. Your team holds final sign-off on every output.
- Your environment, your data — engines run inside infrastructure you approve; outputs and workpapers are yours.

6.0 ENGAGEMENT

Engines are built custom for your company against your ledgers, your close rhythm, and your review standards — the portfolio above is the proof of capability, not a boxed product. A first conversation costs nothing: you describe the work that consumes your team's month; we tell you plainly what an engine can take over, what it cannot, and what a scoped first phase would cost. Stop after any phase; everything built to that point is yours.

SOPHON FINANCE SYSTEMS

Sophon Narith, Founder
contact@sophonfinance.com
sophonfinance.com

linkedin.com/in/sophonnarith
github.com/sophonfinance-wq
Seattle, Washington

Disclosures: Human-hour baselines and dollar figures are engineering estimates for the equivalent scope, not measurements of any client's staff. Public demonstrations use fictional, seeded data; no client information appears in public code or in this report. Sophon Finance Systems provides automation engineering; final review and filing authority remain with your team and your licensed professionals.